4

Differentiation

4.1 The Lebesgue-Radon-Nikodym theorem

Definition 4.1.1: Let $(\Omega, \mathcal{F})$ be a measurable space and let μ and ν be two measures on $(\Omega, \mathcal{F})$. The measure μ is said to be *dominated by* ν or *absolutely continuous w.r.t.* ν and written as $\mu \ll \nu$ if

$$\nu(A) = 0 \Rightarrow \mu(A) = 0 \quad \text{for all} \quad A \in \mathcal{F}. \tag{1.1}$$

Example 4.1.1: Let m be the Legesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and let μ be the standard normal distribution, i.e.,

$$\mu(A) \equiv \int_A \frac{1}{\sqrt{2\pi}} e^{-x^2/2} m(dx), \quad A \in \mathcal{B}(\mathbb{R}).$$

Then $m(A) = 0 \Rightarrow \mu(A) = 0$ and hence $\mu \ll m$.

Example 4.1.2: Let $\mathbb{Z}_+ \equiv \{0, 1, 2, \ldots\}$ denote the set of all nonnegative integers. Let ν be the counting measure on $\Omega = \mathbb{Z}_+$ and μ be the Poisson (λ) distribution for $0 < \lambda < \infty$, i.e.,

$$\nu(A) = \text{number of elements in } A$$

and

$$\mu(A) = \sum_{j \in A} \frac{e^{-\lambda} \lambda^j}{j!}$$

for all $A \in \mathcal{P}(\Omega)$, the power set of Ω. Since $\nu(A) = 0 \Leftrightarrow A = \emptyset \Leftrightarrow \mu(A) = 0$, it follows that $\mu \ll \nu$ and $\nu \ll \mu$.

Example 4.1.3: Let f be a nonnegative measurable function on a measure space $(\Omega, \mathcal{F}, \nu)$. Let

$$\mu(A) \equiv \int_A f d\nu \quad \text{for all} \quad A \in \mathcal{F}. \tag{1.2}$$

Then, μ is a measure on $(\Omega, \mathcal{F})$ and $\nu(A) = 0 \Rightarrow \mu(A) = 0$ for all $A \in \mathcal{F}$ and hence $\mu \ll \nu$.

The Radon-Nikodym theorem is a sort of converse to Example 4.1.3. It says that if μ and ν are σ-finite measures (see Section 1.2) on a measurable space $(\Omega, \mathcal{F})$ and if $\mu \ll \nu$, then there is a nonnegative measurable function f on $(\Omega, \mathcal{F})$ such that (1.2) holds.

Definition 4.1.2: Let $(\Omega, \mathcal{F})$ be a measurable space and let μ and ν be two measures on $(\Omega, \mathcal{F})$. Then, μ is called *singular* w.r.t. ν and written as $\mu \perp \nu$ if there exists a set $B \in \mathcal{F}$ such that

$$\mu(B) = 0 \quad \text{and} \quad \nu(B^c) = 0. \tag{1.3}$$

Note that μ is singular w.r.t. ν implies that ν is singular w.r.t. μ. Thus, the notion of singularity between two measures μ and ν is symmetric but that of absolutely continuity is not. Note also that if μ and ν are mutually singular and B satisfies (1.3), then for all $A \in \mathcal{F}$,

$$\mu(A) = \mu(A \cap B^c) \quad \text{and} \quad \nu(A) = \nu(A \cap B). \tag{1.4}$$

Example 4.1.4: Let μ be the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and ν be defined as $\nu(A) = \#$ elements in $A \cap \mathbb{Z}$ where $\mathbb{Z}$ is the set of integers. Then

$$\nu(\mathbb{Z}^c) = 0 \quad \text{and} \quad \mu(\mathbb{Z}) = 0$$

and hence (1.3) holds with $B = \mathbb{Z}$. Thus μ and ν are *mutually singular*.

Another example is the pair m and μ_c on [0,1] where μ_c is the Lebesgue-Stieltjes measure generated by the Cantor function (cf. Section 4.5) and m is the Lebesgue measure.

Example 4.1.5: Let μ be the Lebesgue measure restricted to $(-\infty, 0]$ and ν be the Exponential(1) distribution. That is, for any $A \in \mathcal{B}(\mathbb{R})$,

$$\begin{aligned} \mu(A) &= \text{the Lebesgue measure of} \quad A \cap (-\infty, 0]; \\ \nu(A) &= \int_{A \cap (0,\infty)} e^{-x} dx. \end{aligned}$$

Then, $\mu((0,\infty)) = 0$ and $\nu((-\infty, 0]) = 0$, and (1.3) holds with $B = (-\infty, 0]$.

Suppose that μ and ν are two finite measures on a measurable space $(\Omega, \mathcal{F})$. H. Lebesgue showed that μ_1 can be decomposed as a sum of two measures, i.e.,

$$\mu = \mu_a + \mu_s$$

where $\mu_a \ll \nu$ and $\mu_s \perp \nu$. The next theorem is the main result of this section and it combines the above decomposition result of Lebesgue with the Radon-Nikodym theorem mentioned earlier.

Theorem 4.1.1: *Let $(\Omega, \mathcal{F})$ be a measurable space and let μ_1 and μ_2 be two σ-finite measures on $(\Omega, \mathcal{F})$.*

(i) (The Lebesgue decomposition theorem). The measure μ_1 can be uniquely decomposed as

$$\mu_1 = \mu_{1a} + \mu_{1s} \tag{1.5}$$

where μ_{1a} and μ_{1s} are σ-finite measures on $(\Omega, \mathcal{F})$ such that $\mu_{1a} \ll \mu_2$ and $\mu_{1s} \perp \mu_2$.

(ii) (The Radon-Nikodym theorem). There exists a nonnegative measurable function h on $(\Omega, \mathcal{F})$ such that

$$\mu_{1a}(A) = \int_A h d\mu_2 \quad \textit{for all} \quad A \in \mathcal{F}. \tag{1.6}$$

Proof: Case 1: Suppose that μ_1 and μ_2 are finite measures. Let μ be the measure $\mu = \mu_1 + \mu_2$ and let $H = L^2(\mu)$. Define a linear function T on H by

$$T(f) = \int f d\mu_1. \tag{1.7}$$

Then, by the Cauchy-Schwarz inequality applied to the functions f and $g \equiv 1$,

$$\begin{aligned} |T(f)| &\leq \Big(\int f^2 d\mu_1\Big)^{\frac{1}{2}} \Big(\mu_1(\Omega)\Big)^{\frac{1}{2}} \\ &\leq \Big(\int f^2 d\mu\Big)^{\frac{1}{2}} \Big(\mu_1(\Omega)\Big)^{\frac{1}{2}}. \end{aligned}$$

This shows that T is a bounded linear functional on H with $\|T\| \leq M \equiv (\mu_1(\Omega))^{\frac{1}{2}}$. By the Riesz representation theorem (cf. Theorem 3.3.3 and Remark 3.3.2), there exists a $g \in L^2(\mu)$ such that

$$T(f) = \int f g d\mu \tag{1.8}$$

for all $f \in L^2(\mu)$. Let $f = I_A$ for A in $\mathcal{F}$. Then, (1.7) and (1.8) yield

$$\mu_1(A) = T(I_A) = \int_A g d\mu.$$

But $0 \le \mu_1(A) \le \mu(A)$ for all $A \in \mathcal{F}$. Hence the function g in $L^2(\mu)$ satisfies

$$0 \le \int_A g d\mu \le \mu(A) \quad \text{for all} \quad A \in \mathcal{F}. \tag{1.9}$$

Let $A_1 = \{0 \le g < 1\}$, $A_2 = \{g = 1\}$, $A_3 = \{g \notin [0,1]\}$. Then (1.9) implies that $\mu(A_3) = 0$ (see Problem 4.1). Now define the measures $\mu_{1a}(\cdot)$ and $\mu_{1s}(\cdot)$ by

$$\mu_{1a}(A) \equiv \mu_1(A \cap A_1), \quad \mu_{1s}(A) \equiv \mu_1(A \cap A_2), \quad A \in \mathcal{F}. \tag{1.10}$$

Next it will be shown that $\mu_{1a} \ll \mu_2$ and $\mu_{1s} \perp \mu_2$, thus establishing (1.5). By (1.7) and (1.8), for all $f \in H$,

$$\int f d\mu_1 = \int fg d\mu = \int fg d\mu_1 + \int fg d\mu_2$$

$$\Rightarrow \int f(1-g) d\mu_1 = \int fg d\mu_2. \tag{1.11}$$

Setting $f = I_{A_2}$ yields

$$0 = \mu_2(A_2).$$

From (1.10), since $\mu_{1s}(A_2^c) = 0$, it follows that $\mu_{1s} \perp \mu_2$. Now fix $n \ge 1$ and $A \in \mathcal{F}$. Let $f = I_{A \cap A_1}(1 + g + \ldots + g^{n-1})$. Then (1.11) implies that

$$\int_{A \cap A_1} (1 - g^n) d\mu_1 = \int_{A \cap A_1} g(1 + g + \ldots + g^{n-1}) d\mu_2.$$

Now letting $n \to \infty$, and using the MCT on both sides, yields

$$\mu_{1a}(A) = \int_A I_{A_1} \frac{g}{(1-g)} d\mu_2. \tag{1.12}$$

Setting $h \equiv \frac{g}{1-g} I_{A_1}$ completes the proof of (1.5) and (1.6).

CASE 2: Now suppose that μ_1 and μ_2 are σ-finite. Then there exists a countable partition $\{D_n\}_{\ge 1} \subset \mathcal{F}$ of Ω such that $\mu_1(D_n)$ and $\mu_2(D_n)$ are both finite for all $n \ge 1$. Let $\mu_1^{(n)}(\cdot) \equiv \mu_1(\cdot \cap D_n)$ and $\mu_2^{(n)} \equiv \mu_2(\cdot \cap D_n)$. Then applying 'Case 1' to $\mu_1^{(n)}$ and $\mu_2^{(n)}$ for each $n \ge 1$, one gets measures $\mu_{1a}^{(n)}, \mu_{1s}^{(n)}$ and a function h_n such that

$$\mu_1^{(n)}(\cdot) \equiv \mu_{1a}^{(n)}(\cdot) + \mu_{1s}^{(n)}(\cdot) \tag{1.13}$$

where, for A in $\mathcal{F}$, $\mu_{1a}^{(n)}(A) = \int_A h_n d\mu_2^{(n)} = \int_A h_n I_{D_n} d\mu_2$ and $\mu_{1s}^{(n)}(\cdot) \perp \mu_2^{(n)}$. Since $\mu_1(\cdot) = \sum_{n=1}^{\infty} \mu_1^{(n)}(\cdot)$, it follows from (1.13) that

$$\mu_1(\cdot) = \mu_{1a}(\cdot) + \mu_{1s}(\cdot), \tag{1.14}$$

where $\mu_{1a}(A) \equiv \sum_{n=1}^{\infty} \mu_{1a}^{(n)}(A)$ and $\mu_{1s}(\cdot) = \sum_{n=1}^{\infty} \mu_{1s}^{(n)}(\cdot)$. By the MCT,

$$\mu_{1a}(A) = \int_A h d\mu_2, \quad A \in \mathcal{F},$$

where $h \equiv \sum_{n=1}^{\infty} h_n I_{D_n}$.

Clearly, $\mu_{1a} \ll \mu_2$. The verification of the singularity of μ_{1s} and μ_2 is left as an exercise (Problem 4.2).

It remains to prove the uniqueness of the decomposition. Let $\mu_1 = \mu_a + \mu_s$ and $\mu_1 = \mu_a' + \mu_s'$ be two decompositions of μ_1 where μ_a and μ_a' are absolutely continuous w.r.t. μ_2 and μ_s and μ_s' are singular w.r.t. μ_2. By definition, there exist sets B and B' in $\mathcal{F}$ such that

$$\mu_2(B) = 0, \mu_2(B') = 0, \quad \text{and} \quad \mu_s(B^c) = 0, \mu_s'(B'^c) = 0.$$

Let $D = B \cup B'$. Then $\mu_2(D) = 0$ and $\mu_s(D^c) \le \mu_s(B^c) = 0$. Similarly, $\mu_s'(D^c) \le \mu_s'(B'^c) = 0$. Also $\mu_2(D) = 0$ implies $\mu_a(D) = 0 = \mu_a'(D)$. Thus for any $A \in \mathcal{F}$,

$$\mu_a(A) = \mu_a(A \cap D^c) \quad \text{and} \quad \mu_a'(A) = \mu_a'(A \cap D^c).$$

Also

$$\begin{aligned} \mu_s(A \cap D^c) &\le \mu_s(A \cap B^c) = 0 \\ \mu_s'(A \cap D^c) &\le \mu_s'(A \cap B'^c) = 0. \end{aligned}$$

Thus, $\mu(A \cap D^c) = \mu_a(A \cap D^c) + \mu_s(A \cap D^c) = \mu_a(A \cap D^c) = \mu_a(A)$ and $\mu(A \cap D^c) = \mu_a'(A \cap D^c) + \mu_s'(A \cap D^c) = \mu_a'(A \cap D^c) = \mu_a'(A)$. Hence, $\mu_a(A) = \mu(A \cap D^c) = \mu_a'(A)$ for every $A \in \mathcal{F}$. That is, $\mu_a = \mu_a'$ and hence, $\mu_s = \mu_s'$. $\square$

Remark 4.1.1: In Theorem 4.1.1, the hypothesis of σ finiteness cannot be dropped. For example, let μ be the Lebesgue measure and ν be the counting measure on $[0, 1]$. Then $\mu \ll \nu$ but there does not exist a nonnegative $\mathcal{F}$-measurable function h such that $\mu(A) = \int_A h d\nu$. To see this, if possible, suppose that for some $h \in L^1(\nu)$, $\mu(A) = \int_A h d\nu$ for all $A \in \mathcal{F}$. Note that $\mu([0, 1]) = 1$ implies that $\int_{[0,1]} h d\nu < \infty$ and hence, that $B \equiv \{x : h(x) > 0\}$ is countable (Problem 4.3). But μ being the Lebesgue measure, $\mu(B) = 0$ and $\mu(B^c) = 1$. Since by definition, $h \equiv 0$ on B^c, this implies $1 = \mu(B^c) = \int_{B^c} h d\nu = 0$, leading to a contradiction. However, if ν is σ-finite and $\mu \ll \nu$ (μ not necessarily σ-finite), then the Radon-Nikodym theorem holds, i.e., *there exists a nonnegative $\mathcal{F}$-measurable function h such that*

$$\mu(A) = \int_A h d\nu \quad \textit{for all} \quad A \in \mathcal{F}.$$

For a proof, see Royden (1988), Chapter 11.

Definition 4.1.3: Let μ and ν be measures on a measurable space $(\Omega, \mathcal{F})$ and let h be a nonnegative measurable function such that

$$\mu(A) = \int_A h d\nu \quad \text{for all} \quad A \in \mathcal{F}.$$

Then h is called the *Radon-Nikodym derivative* of μ w.r.t. ν and is written as

$$\frac{d\mu}{d\nu} = h.$$

If $\mu(\Omega) < \infty$ and there exist two nonnegative $\mathcal{F}$-measurable functions h_1 and h_2 such that

$$\mu(A) = \int_A h_1 d\nu = \int_A h_2 d\nu$$

for all $A \in \mathcal{F}$, then $h_1 = h_2$ a.e. (ν) and thus the Radon-Nikodym derivative $\frac{d\mu}{d\nu}$ is unique up to equivalence a.e. (ν). This also extends to the case when μ is σ-finite.

The following proposition is easy to verify (cf. Problem 4.4).

Proposition 4.1.2: *Let $\nu, \mu, \mu_1, \mu_2, \ldots$ be σ-finite measures on a measurable space $(\Omega, \mathcal{F})$.*

(i) If $\mu_1 \ll \mu_2$ and $\mu_2 \ll \mu_3$, then $\mu_1 \ll \mu_3$ and

$$\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2}\frac{d\mu_2}{d\mu_3} \quad a.e.\ (\mu_3).$$

(ii) Suppose that μ_1 and μ_2 are dominated by μ_3. Then for any $\alpha, \beta \geq 0$, $\alpha\mu_1 + \beta\mu_2$ is dominated by μ_3 and

$$\frac{d(\alpha\mu_1 + \beta\mu_2)}{d\mu_3} = \alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3} \quad a.e.\ (\mu_3).$$

(iii) If $\mu \ll \nu$ and $\frac{d\mu}{d\nu} > 0$ a.e. (ν), then $\nu \ll \mu$ and

$$\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1} \quad a.e.\ (\mu).$$

(iv) Let $\{\mu_n\}_{n\geq 1}$ be a sequence of measures and $\{\alpha_n\}_{n\geq 1}$ be a sequence of positive real numbers, i.e., $\alpha_n > 0$ for all $n \geq 1$. Define $\mu = \sum_{n=1}^{\infty} \alpha_n \mu_n$.

(a) Then, $\mu \ll \nu$ iff $\mu_n \ll \nu$ for each $n \geq 1$ and in this case,

$$\frac{d\mu}{d\nu} = \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu} \quad a.e.\ (\nu).$$

(b) $\mu \perp \nu$ iff $\mu_n \perp \nu$ for all $n \geq 1$.

4.2 Signed measures

Let μ_1 and μ_2 be two finite measures on a measurable space $(\Omega, \mathcal{F})$. Let

$$\nu(A) \equiv \mu_1(A) - \mu_2(A), \quad \text{for all} \quad A \in \mathcal{F}. \tag{2.1}$$

Then $\nu : F \to \mathbb{R}$ satisfies the following:

(i) $\nu(\emptyset) = 0$.

(ii) For any $\{A_n\}_{n \geq 1} \subset \mathcal{F}$ with $A_i \cap A_j = \emptyset$ for $i \neq j$, and with $\sum_{i=1}^{\infty} |\nu(A_i)| < \infty$,

$$\nu(A) = \sum_{i=1}^{\infty} \nu(A_i). \tag{2.2}$$

(iii) Let

$$\|\nu\| \equiv \sup\Big\{ \sum_{i=1}^{\infty} |\nu(A_i)| : \{A_n\}_{n \geq 1} \subset \mathcal{F},\ A_i \cap A_j = \emptyset \quad \text{for} \quad i \neq j,\ \bigcup_{n \geq 1} A_n = \Omega \Big\}. \tag{2.3}$$

Then, $\|\nu\|$ is finite.

Note that (iii) holds because $\|\nu\| \leq \mu_1(\Omega) + \mu_2(\Omega) < \infty$.

Definition 4.2.1: A set function $\nu : \mathcal{F} \to \mathbb{R}$ satisfying (i), (ii), and (iii) above is called a *finite signed measure.*

The above example shows that the difference of two finite measures is a finite signed measure. It will be shown below that every finite signed measure can be expressed as the difference of two finite measures.

Proposition 4.2.1: *Let ν be a finite signed measure on $(\Omega, \mathcal{F})$. Let*

$$|\nu|(A) \equiv \sup\Big\{ \sum_{n=1}^{\infty} |\nu(A_n)| : \{A_n\}_{n \geq 1} \subset \mathcal{F}, A_i \cap A_j = \emptyset \quad \text{for} \quad i \neq j,\ \bigcup_{n \geq 1} A_n = A \Big\}. \tag{2.4}$$

Then $|\nu|(\cdot)$ *is a finite measure on* $(\Omega, \mathcal{F})$.

Proof: That $|\nu|(\Omega) < \infty$ follows from part (iii) of the definition. Thus it is enough to verify that $|\nu|$ is countably additive. Let $\{A_n\}_{n\geq 1}$ be a countable family of disjoint sets in $\mathcal{F}$. Let $A = \bigcup_{n\geq 1} A_n$. By the definition of $|\nu|$, for all $\epsilon > 0$ and $n \in \mathbb{N}$, there exists a countable family $\{A_{nj}\}_{j\geq 1}$ of disjoint sets in $\mathcal{F}$ with $A_n = \bigcup_{j\geq 1} A_{nj}$ such that $\sum_{j=1}^{\infty} |\nu(A_{nj})| > |\nu|(A_n) - \frac{\epsilon}{2^n}$. Hence,

$$\sum_{n=1}^{\infty}\sum_{j=1}^{\infty} |\nu(A_{nj})| > \sum_{n=1}^{\infty} |\nu|(A_n) - \epsilon.$$

Note that $\{A_{nj}\}_{n\geq 1, j\geq 1}$ is a countable family of disjoint sets in $\mathcal{F}$ such that $A = \bigcup_{n\geq 1} A_n = \bigcup_{n\geq 1}\bigcup_{j\geq 1} A_{nj}$. It follows from the definition of $|\nu|$ that

$$|\nu|(A) \geq \sum_{n=1}^{\infty}\sum_{j=1}^{\infty} |\nu(A_{nj})| > \sum_{n=1}^{\infty} |\nu|(A_n) - \epsilon.$$

Since this is true for for all $\epsilon > 0$, it follows that

$$|\nu|(A) \geq \sum_{n=1}^{\infty} |\nu|(A_n). \tag{2.5}$$

To get the opposite inequality, let $\{B_j\}_{j\geq 1}$ be a countable family of disjoint sets in $\mathcal{F}$ such that $\bigcup_{j\geq 1} B_j = A = \bigcup_{n\geq 1} A_n$. Since $B_j = B_j \cap A = \bigcup_{n\geq 1}(B_j \cap A_n)$ and ν satisfies (2.2),

$$\nu(B_j) = \sum_{n=1}^{\infty} \nu(B_j \cap A_n) \quad \text{for all} \quad j \geq 1.$$

Thus

$$\begin{aligned} \sum_{j=1}^{\infty} |\nu(B_j)| &\leq \sum_{j=1}^{\infty}\sum_{n=1}^{\infty} |\nu(B_j \cap A_n)| \\ &= \sum_{n=1}^{\infty}\sum_{j=1}^{\infty} |\nu(B_j \cap A_n)|. \end{aligned} \tag{2.6}$$

Note that for each A_n, $\{B_j \cap A_n\}_{j\geq 1}$ is a countable family of disjoint sets in $\mathcal{F}$ such that $A_n = \bigcup_{j\geq 1}(B_j \cap A_n)$. Hence from (2.4), it follows that $|\nu|(A_n) \geq \sum_{j=1}^{\infty} |\nu(B_j \cap A_n)|$ and hence, $\sum_{n=1}^{\infty} |\nu|(A_n) \geq \sum_{n=1}^{\infty}\sum_{j=1}^{\infty} |\nu(B_j \cap A_n)|$. From (2.6), it follows that $\sum_{n=1}^{\infty} |\nu|(A_n) \geq \sum_{j=1}^{\infty} |\nu(B_j)|$. This being true for every such family $\{B_j\}_{j\geq 1}$, it follows from (2.4) that

$$|\nu|(A) \leq \sum_{i=1}^{\infty} |\nu|(A_i) \tag{2.7}$$

and with (2.5), this completes the proof. □

Definition 4.2.2: The measure $|\nu|$ defined by (2.4) is called the *total variation measure* of the signed measure ν.

Next, define the set functions

$$\nu^+ \equiv \frac{|\nu| + \nu}{2}, \quad \nu^- \equiv \frac{|\nu| - \nu}{2}. \tag{2.8}$$

It can be verified that ν^+ and ν^- are both finite measures on $(\Omega, \mathcal{F})$.

Definition 4.2.3: The measures ν^+ and ν^- are called the *positive* and *negative variation measures* of the signed measure ν, respectively.

It follows from (2.8) that

$$\nu = \nu^+ - \nu^-. \tag{2.9}$$

Thus every finite signed measure ν on $(\Omega, \mathcal{F})$ is the difference of two finite measures, as claimed earlier.

Note that both ν^+ and ν^- are dominated by $|\nu|$ and all three measures are finite. By the Radon-Nikodym theorem (Theorem 4.1.1), there exist functions h_1 and h_2 in $L^1(\Omega, \mathcal{F}, |\nu|)$ such that

$$\frac{d\nu^+}{d|\nu|} = h_1 \quad \text{and} \quad \frac{d\nu^-}{d|\nu|} = h_2. \tag{2.10}$$

This and (2.9) imply that for any A in $\mathcal{F}$,

$$\nu(A) = \int_A h_1 d|\nu| - \int_A h_2 d|\nu| = \int_A h d|\nu|, \tag{2.11}$$

where $h = h_1 - h_2$. Thus every finite signed measure ν on $(\Omega, \mathcal{F})$ can be expressed as

$$\nu(A) = \int_A f d\mu, \ A \in \mathcal{F} \tag{2.12}$$

for some finite measure μ on $(\Omega, \mathcal{F})$ and some $f \in L^1(\Omega, \mathcal{F}, \mu)$.

Conversely, it is easy to verify that a set function ν defined by (2.12) for some finite measure μ on $(\Omega, \mathcal{F})$ and some $f \in L^1(\Omega, \mathcal{F}, \mu)$ is a finite signed measure (cf. Problem 4.6). This leads to the following:

Theorem 4.2.2:

(i) A set function ν on a measurable space $(\Omega, \mathcal{F})$ is a finite signed measure iff there exist two finite measures μ_1 and μ_2 on $(\Omega, \mathcal{F})$ such that $\nu = \mu_1 - \mu_2$.

(ii) A set function ν on a measurable space $(\Omega, \mathcal{F})$ is a finite signed measure iff there exist a finite measure μ on $(\Omega, \mathcal{F})$ and an $f \in L^1(\Omega, \mathcal{F}, \mu)$ such that for all A in $\mathcal{F}$,

$$\nu(A) = \int_A f \, d\mu.$$

Definition 4.2.4: Let ν be a finite signed measure on a measurable space on $(\Omega, \mathcal{F})$. A set $A \in \mathcal{F}$ is called a *positive set* for ν if for any $B \subset A, B \in \mathcal{F}$, $\nu(B) \geq 0$. A set $A \in \mathcal{F}$ is called a *negative set* for ν if for any $B \subset A$ with $B \in \mathcal{F}$, $\nu(B) \leq 0$.

Let h be as in (2.11). Let

$$\Omega^+ = \{\omega : h(\omega) \geq 0\} \quad \text{and} \quad \Omega^- = \{\omega : h(\omega) < 0\}. \tag{2.13}$$

From (2.11), it follows that for all A in $\mathcal{F}$, $\nu(A \cap \Omega^+) \geq 0$ and $\nu(A \cap \Omega^-) \leq 0$. Thus Ω^+ is a positive set and Ω^- is a negative set for ν. Furthermore, $\Omega^+ \cup \Omega^- = \Omega$ and $\Omega^+ \cap \Omega^- = \emptyset$. Summarizing this discussion, one gets the following theorem.

Theorem 4.2.3: (*Hahn decomposition theorem*). *Let ν be a finite signed measure on a measurable space $(\Omega, \mathcal{F})$. Then there exist a positive set Ω^+ and a negative set Ω^- for ν such that $\Omega = \Omega^+ \cup \Omega^-$ and $\Omega^+ \cap \Omega^- = \emptyset$.*

Let Ω^+ and Ω^- be as in (2.13). It can be verified (Problem 4.8) that for any $B \in \mathcal{F}$, if $B \subset \Omega^+$, then $\nu(B) = |\nu|(B)$. By (2.11), this implies that for all A in $\mathcal{F}$,

$$\int_{A \cap \Omega^+} h d|\nu| = |\nu|(A \cap \Omega^+).$$

It follows that $h = 1$ a.e. $(|\nu|)$ on Ω^+. Similarly, $h = -1$ a.e. $(|\nu|)$ on Ω^-. Thus, the measures ν^+ and ν^-, defined in (2.8), reduce to

$$\begin{aligned} \nu^+(A) &= \int_A \frac{(1+h)}{2} d|\nu| \\ &= \int_{A \cap \Omega^+} \frac{(1+h)}{2} d|\nu| + \int_{A \cap \Omega^-} \frac{(1+h)}{d} |\nu| \\ &= |\nu|(A \cap \Omega^+), \end{aligned}$$

and similarly

$$\nu^-(A) = |\nu|(A \cap \Omega^-).$$

Note that ν^+ and ν^- are both finite measures that are mutually singular. This particular decomposition of ν as

$$\nu = \nu^+ - \nu^-$$

is known as the *Jordan decomposition* of ν. It will now be shown that this decomposition is minimal and that it is unique in the class of signed measures with mutually singular components. Suppose there exist two finite measures μ_1 and μ_2 on $(\Omega, \mathcal{F})$ such that $\nu = \mu_1 - \mu_2$. For any $A \in \mathcal{F}$, $\nu^+(A) = \nu(A \cap \Omega^+) = \mu_1(A \cap \Omega^+) - \mu_2(A \cap \Omega^+) \leq \mu_1(A \cap \Omega^+) \leq \mu_1(A)$ and $\nu^-(A) = -\nu(A \cap \Omega^-) = \mu_2(A \cap \Omega^-) - \mu_1(A \cap \Omega^-) \leq \mu_2(A \cap \Omega^-) \leq \mu_2(A)$. Thus, $\nu^+ \leq \mu_1$ and $\nu^- \leq \mu_2$. Clearly, since both μ_1 and ν^+ are finite measures on $(\Omega, \mathcal{F})$, $\mu_1 - \nu^+$ is also a finite measure. Similarly, $\mu_2 - \nu^-$ is also a finite measure. Also, since $\mu_1 - \mu_2 = \nu = \nu^+ - \nu^-$, it follows that $\mu_1 - \nu^+ = \mu_2 - \nu^- = \lambda$, say. Thus, for any decomposition of ν as $\mu_1 - \mu_2$ with μ_1, μ_2 finite measures, it holds that $\mu_1 = \nu^+ + \lambda$ and $\mu_2 = \nu^- + \lambda$, where λ is a measure on $(\Omega, \mathcal{F})$. Thus, $\nu = \nu^+ - \nu^-$ is a *minimal* decomposition in the sense that in this case $\lambda = 0$. Now suppose μ_1 and μ_2 are mutually singular, i.e., there exist $\Omega_1, \Omega_2 \in \mathcal{F}$ such that $\Omega_1 \cap \Omega_2 = \emptyset$, $\Omega_1 \cup \Omega_2 = \Omega$, and $\mu_1(\Omega_2) = 0 = \mu_2(\Omega_1)$. Since $\mu_1 \geq \lambda$ and $\mu_2 \geq \lambda$, it follows that $\lambda(\Omega_2) = 0 = \lambda(\Omega_1)$. Thus $\lambda = 0$ and $\mu_1 = \nu^+$ and $\mu_2 = \nu^-$.

Summarizing the above discussion yields:

Theorem 4.2.4: *Let ν be a finite signed measure on a measurable space $(\Omega, \mathcal{F})$ and let μ_1 and μ_2 be two finite measures on $(\Omega, \mathcal{F})$ such that $\nu = \mu_1 - \mu_2$. Then there exists a finite measure λ such that $\mu_1 = \nu^+ + \lambda$ and $\mu_2 = \nu^- + \lambda$ with $\lambda = 0$ iff μ_1 and μ_2 are mutually singular.*

Let

$$\mathbb{S} \equiv \{\nu : \nu \quad \text{is a finite signed measure on} \quad (\Omega, \mathcal{F})\}.$$

Also, for any $\alpha \in \mathbb{R}$, let $\alpha^+ = \max(\alpha, 0)$ and $\alpha^- = \max(-\alpha, 0)$. Note that for ν_1, ν_2 in $\mathbb{S}$ and $\alpha_1, \alpha_2 \in \mathbb{R}$,

$$\begin{aligned}
\alpha_1\nu_1 + \alpha_2\nu_2 &= (\alpha_1^+ - \alpha_1^-)(\nu_1^+ - \nu_1^-) + (\alpha_2^+ - \alpha_2^-)(\nu_2^+ - \nu_2^-) \\
&= (\alpha_1^+\nu_1^+ + \alpha_1^-\nu_1^- + \alpha_2^+\nu_2^+ + \alpha_2^-\nu_2^-) \\
&\quad - (\alpha_1^+\nu_1^- + \alpha_1^-\nu_1^+ + \alpha_2^+\nu_2^- + \alpha_2^-\nu_2^+) \\
&= \lambda_1 - \lambda_2, \quad \text{say,}
\end{aligned}$$

where λ_1 and λ_2 are both finite measures. It now follows from Theorem 4.2.2 that $\alpha_1\nu_1 + \alpha_2\nu_2 \in \mathbb{S}$. Thus, $\mathbb{S}$ is a vector space over $\mathbb{R}$.

Now it will be shown that $\|\nu\| \equiv |\nu|(\Omega)$ is a norm on $\mathbb{S}$ and that $(\mathbb{S}, \|\cdot\|)$ is a Banach space.

Definition 4.2.5: For a finite signed measure ν on a measurable space $(\Omega, \mathcal{F})$, the *total variation norm* ν is defined by $\|\nu\| \equiv |\nu|(\Omega)$.

Proposition 4.2.5: *Let $\mathbb{S} \equiv \{\nu : \nu$ a finite signed measure on $(\Omega, \mathcal{F})\}$. Then, $\|\nu\| \equiv |\nu|(\Omega)$, $\nu \in \mathbb{S}$ is a norm on $\mathbb{S}$.*

Proof: Let $\nu_1, \nu_2 \in \mathbb{S}$, $\alpha_1, \alpha_2 \in \mathbb{R}$ and $\lambda = \alpha_1\nu_1 + \alpha_2\nu_2$. For any $A \in \mathcal{F}$ and any $\{A_i\}_{i\geq 1} \subset \mathcal{F}$ with $A = \bigcup_{i\geq 1} A_i$,

$$\begin{aligned}
& |\lambda(A_i)| \leq |\alpha_1||\nu_1(A_i)| + |\alpha_2||\nu_2(A_i)| \quad \text{for all} \quad i \geq 1 \\
\Rightarrow \quad & \sum_{i\geq 1} |\lambda(A_i)| \leq |\alpha_1| \sum_{i\geq 1} |\nu_1(A_i)| + |\alpha_2| \sum_{i\geq 1} |\nu_2(A_i)| \\
& \qquad \leq |\alpha_1||\nu_1|(A) + |\alpha_2||\nu_2|(A).
\end{aligned}$$

Taking supremum over all $\{A_i\}_{i\geq 1}$ yields,

$$\begin{aligned}
& |\lambda|(A) \leq |\alpha_1||\nu_1|(A) + |\alpha_2||\nu_2|(\cdot) \\
\text{i.e.,} \qquad & |\lambda|(\cdot) \leq |\alpha_1| \; |\nu_1|(\cdot) + |\alpha_2||\nu_2|(\cdot), \\
\Rightarrow \quad & \|\lambda\| \equiv |\lambda|(\Omega) \leq |\alpha_1| \; |\nu_1|(\Omega) + |\alpha_2| \; |\nu_2|(\Omega) \\
& \qquad = |\alpha_1|\|\nu_1\| + |\alpha_2|\|\nu_2\|.
\end{aligned}$$

Taking $\alpha_1 = \alpha_2 = 1$ yields

$$\|\nu_1 + \nu_2\| \leq \|\nu_1\| + \|\nu_2\|,$$

i.e., the triangle inequality holds.

Next taking $\alpha_2 = 0$ yields $\|\alpha_1\nu_1\| \leq |\alpha_1|\|\nu_1\|$. To get the opposite inequality, note that for $\alpha_1 \neq 0$, $\nu_1 = \frac{1}{\alpha_1}\alpha_1\nu_1$ and so $\|\nu_1\| \leq |\frac{1}{\alpha_1}|\|\alpha_1\nu_1\|$. Hence, $|\alpha_1| \, \|\nu_1\| \leq \|\alpha_1\nu_1\|$. Thus, for any $\alpha_1 \neq 0$, $\|\alpha_1\nu\| = |\alpha_1|\|\nu\|$. Finally, $\|\nu\| = 0 \Rightarrow |\nu|(\Omega) = 0 \Rightarrow |\nu|(A) = 0 \quad \text{for all} \quad A \in \mathcal{F} \Rightarrow \nu(A) = 0 \quad \text{for all} \quad A \in \mathcal{F}$, i.e., ν is the zero measure. Thus $\|\cdot\|$ is a norm on $\mathbb{S}$. $\Box$

Proposition 4.2.6: $(\mathbb{S}, \|\cdot\|)$ *is complete.*

Proof: Let $\{\nu_n\}_{n\geq 1}$ be a Cauchy sequence in $(\mathbb{S}, \|\cdot\|)$. Note that for each $A \in \mathcal{F}$, $|\nu_n(A) - \nu_m(A)| \leq |\nu_n - \nu_m|(A) \leq \|\nu_n - \nu_m\|$. Hence, for each $A \in \mathcal{F}$, $\{\nu_n(A)\}_{n\geq 1}$ is a Cauchy sequence in $\mathbb{R}$ and hence

$$\nu(A) \equiv \lim_{n\to\infty} \nu_n(A) \qquad \text{exists.}$$

It will be shown that $\nu(\cdot)$ is a finite signed measure and $\|\nu_n - \nu\| \to 0$ as $n \to \infty$. Let $\{A_i\}_{i\geq 1} \subset \mathcal{F}$, $A_i \cap A_j = \emptyset$ for $i \neq j$, and $A = \bigcup_{i\geq 1} A_i$. Let $x_n \equiv \{\nu_n(A_i)\}_{i\geq 1}$, $n \geq 1$, and let $x_0 \equiv \{\nu(A_i)\}_{i\geq 1}$. Note that each $x_n \in \ell_1$ where $\ell_1 \equiv \{x : x = \{x_i\}_{i\geq 1} \in \mathbb{R}, \; \sum_{i\geq 1} |x_i| < \infty\}$. For $x \in \ell_1$, let $\|x\|_1 = \sum_{i=1}^{\infty} |x_i|$. Then $\|x_n - x_m\| = \sum_{i\geq 1} |\nu_n(A_i) - \nu_m(A_i)| \leq |\nu_n - \nu_m|(A) \leq |\nu_n - \nu_m|(\Omega) \to 0$ as $n, m \to \infty$. But ℓ_1 is complete under $\|\cdot\|_1$. So there exists $x^* \in \ell_1$ such that $\|x_n - x^*\|_1 \to 0$. Since $x_{ni} \equiv \nu_n(A_i) \to \nu(A_i)$ for all $i \geq 1$, it follows that $x_i^* = \nu(A_i)$ for all $i \geq 1$ and that $\sum_{i=1}^{\infty} |\nu(A_i)| < \infty$. Also, for all $n \geq 1$, $\nu_n(A) = \sum_{i=1}^{\infty} \nu_n(A_i)$. Since $\sum_{i\geq 1} |\nu_n(A_i) - \nu(A_i)| \to 0$, $\nu_n(A) \equiv \sum_{i\geq 1} \nu_n(A_i) \to \sum_{i\geq 1} \nu(A_i)$ as $n \to \infty$. But $\nu_n(A) \to \nu(A)$. Thus,

$\nu(A) = \sum_{i=1}^{\infty} \nu(A_i)$. Also for any countable partition $\{A_i\}_{i\geq 1} \subset \mathcal{F}$ of Ω,

$$\sum_{i=1}^{\infty} |\nu(A_i)| = \lim_{n\to\infty} \sum_{i=1}^{\infty} |\nu_n(A_i)| \leq \lim_{n\to\infty} \|\nu_n\| < \infty.$$

Thus $|\nu|(\Omega) < \infty$ and hence, $\nu \in \mathbb{S}$. Finally,

$$\begin{aligned} \|\nu_n - \nu\| \quad = \quad \sup \Bigg\{ & \sum_{i=1}^{\infty} |\nu_n(A_i) - \nu(A_i)| : \{A_i\}_{i\geq 1} \subset \mathcal{F} \\ & \text{is a disjoint partition of} \quad \Omega \Bigg\}. \end{aligned}$$

But for every countable partition $\{A_i\}_{i\geq 1} \subset \mathcal{F}$,

$$\sum_{i=1}^{\infty} |\nu_n(A_i) - \nu(A_i)| = \lim_{m\to\infty} \sum_{i=1}^{\infty} |\nu_n(A_i) - \nu_m(A_i)| \leq \lim_{m\to\infty} \|\nu_n - \nu_m\|.$$

Thus, $\|\nu_n - \nu\| \leq \lim_{m\to\infty} \|\nu_n - \nu_m\|$ and hence, $\lim_{n\to\infty} \|\nu_n - \nu\| \leq \lim_{n\to\infty} \lim_{m\to\infty} \|\nu_n - \nu_m\| = 0$. Hence, $\nu_n \to \nu$ in $\mathbb{S}$. $\square$

Definition 4.2.6: (*Integration w.r.t. signed measures*). Let μ be a finite signed measure on a measurable space $(\Omega, \mathcal{F})$ and $|\mu|$ be its total variation measure as in Definition 4.2.2. Then, for any $f \in L^1(\Omega, \mathcal{F}, |\mu|)$, $\int f d\mu$ is defined as

$$\int f d\mu = \int f d\mu^+ - \int f d\mu^- ,$$

where μ^+ and μ^- are the positive and negative variations of μ as defined in (2.8).

Proposition 4.2.7: *Let μ be a signed measure on a measurable space $(\Omega, \mathcal{F}, P)$. Let $\mu = \lambda_1 - \lambda_2$ where λ_1 and λ_2 are finite measures. Let $f \in L^1(\Omega, \mathcal{F}, \lambda_1 + \lambda_2)$. Then $f \in L^1(\Omega, \mathcal{F}, |\mu|)$ and*

$$\int f d\mu = \int f d\lambda_1 - \int f d\lambda_2. \tag{2.14}$$

Proof: Left as an exercise (Problem 4.13).

4.3 Functions of bounded variation

From the construction of the Lebesgue-Stieltjes measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ discussed in Chapter 1, it is seen that to every nondecreasing right continuous function $F : \mathbb{R} \to \mathbb{R}$, there is a (Radon) measure μ_F on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that

$\mu_F((a,b]) = F(b) - F(a)$ for all $a < b$ and conversely. If μ_1 and μ_2 are two Radon measures and $\mu = \mu_1 - \mu_2$, let

$$\begin{aligned} G(x) &\equiv \begin{cases} \mu((0,x]) & \text{for} \quad x > 0, \\ -\mu((x,0]) & \text{for} \quad x < 0, \\ 0 & \text{for} \quad x = 0. \end{cases} \\ &= \begin{cases} F_1(x) - F_2(x) - (F_1(0) - F_2(0)) & \text{for} \quad x > 0, \\ (F_1(0) - F_2(0)) - (F_1(x) - F_2(x)) & \text{for} \quad x < 0, \\ 0 & \text{for} \quad x = 0. \end{cases} \end{aligned}$$

Thus to every finite signed measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, there corresponds a function $G(\cdot)$ that is the difference of two right continuous nondecreasing and bounded functions. The converse is also easy to establish. A characterization of such a function $G(\cdot)$ without any reference to measures is given below.

Definition 4.3.1: Let $f : [a,b] \to \mathbb{R}$, where $-\infty < a < b < \infty$. Then for any partition $Q = \{a = x_0 < x_1 < x_2 < \ldots < x_n = b\}$, $n \in \mathbb{N}$, the *positive, negative and total variations of f with respect to Q* are respectively defined as

$$\begin{aligned} P(f,Q) &\equiv \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))^+ \\ N(f,Q) &\equiv \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))^- \\ T(f,Q) &\equiv \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})|. \end{aligned}$$

It is easy to verify that (i) if f is nondecreasing, then

$$P(f,Q) = T(f,Q) = f(b) - f(a) \quad \text{and} \quad N(f,Q) = 0$$

and that (ii) for any f,

$$P(f,Q) + N(f,Q) = T(f,Q).$$

Definition 4.3.2: Let $f = [a,b] \to \mathbb{R}$, where $-\infty < a < b < \infty$. The *positive, negative and total variations of f over $[a,b]$* are respectively defined as

$$\begin{aligned} P(f,[a,b]) &\equiv \sup_Q P(f,Q) \\ N(f,[a,b]) &\equiv \sup_Q N(f,Q) \\ T(f,[a,b]) &\equiv \sup_Q T(f,Q), \end{aligned}$$

where the supremum in each case is taken over all finite partitions Q of $[a,b]$.

Definition 4.3.3: Let $f : [a,b] \to \mathbb{R}$, where $-\infty < a < b < \infty$. Then, f is said to be of *bounded variation on* $[a,b]$ if $T(f,[a,b]) < \infty$. The set of all such functions is denoted by $BV[a,b]$.

As remarked earlier, if f is nondecreasing, then $T(f,Q) = f(b) - f(a)$ for each Q and hence $T(f,[a,b]) = f(b) - f(a)$. It follows that if $f = f_1 - f_2$, where both f_1 and f_2 are nondecreasing, then $f \in BV[a,b]$. A natural question is whether the converse is true. The answer is yes, as shown by the following result.

Theorem 4.3.1: *Let $f \in BV[a,b]$. Let $f_1(x) \equiv P(f,[a,x])$ and $f_2(x) \equiv N(f,[a,x])$. Then f_1 and f_2 are nondecreasing in $[a,b]$ and for all $a \le x \le b$,*

$$f(x) = f_1(x) - f_2(x)$$

Proof: That f_1 and f_2 are nondecreasing follows from the definition. It is enough to verify that if $f \in BV[a,b]$, then

$$f(b) - f(a) = P(f,[a,b]) - N(f,[a,b]),$$

as this can be applied to $[a,x]$ for $a \le x < b$. For each finite partition Q of $[a,b]$,

$$\begin{aligned} f(b) - f(a) &= \sum_{i=1}^{n} (f(x_i) - f(x_{i-1})) \\ &= P(f,Q) - N(f,Q). \end{aligned}$$

Thus $P(f,Q) = f(b) - f(a) + N(f,Q)$. By taking supremum over all finite partitions Q, it follows that

$$P(f,[a,b]) = f(b) - f(a) + N(f,[a,b]).$$

If $f \in BV[a,b]$, this yields $f(b) - f(a) = P(f,[a,b]) - N(f,[a,b])$. $\square$

Remark 4.3.1: Since $T(f,Q) = P(f,Q) + N(f,Q) = 2P(f,Q) - (f(b) - f(a))$, it follows that if $f \in BV[a,b]$, then

$$\begin{aligned} T(f,[a,b]) &= 2P(f,[a,b]) - (f(b) - f(a)) \\ &= P(f,[a,b]) + N(f,[a,b]). \end{aligned}$$

Corollary 4.3.2: *A function $f \in BV[a,b]$ iff there exists a finite signed measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $f(x) = \mu([a,x])$, $a \le x \le b$.*

The proof of this corollary is left as an exercise.

Remark 4.3.2: Some observations on functions of bounded variations are listed below.

(a) Let $f = I_{\mathbb{Q}}$ where $\mathbb{Q}$ is the set of rationals. Then for any $[a,b], a < b, P(f,[a,b]) = N(f,[a,b]) = \infty$ and so $f \notin BV[a,b]$. This holds for $f = I_D$ for any set D such that both D and D^c are dense in $\mathbb{R}$.

(b) Let f be *Lipschitz* on $[a,b]$. That is, $|f(x) - f(y)| \leq K|x-y|$ for all x, y in $[a,b]$ where $K \in (0,\infty)$ is a constant. Then, $f \in BV[a,b]$.

(c) Let f be differentiable in (a,b) and continuous on $[a,b]$ and $f'(\cdot)$ be bounded in (a,b). Then by the mean value theorem, f is Lipschitz and hence, f is in $BV[a,b]$.

(d) Let $f(x) = x^2 \sin \frac{1}{x},\ 0 < x \leq 1$, and let $f(0) = 0$. Then f is continuous on $[0,1]$, differentiable on $(0,1)$ with f' bounded on $(0,1)$, and hence $f \in BV[0,1]$.

(e) Let $g(x) = x^2 \sin \frac{1}{x^2},\ 0 < x \leq 1,\ g(0) = 0$. Then g is continuous on $[0,1]$, differentiable on $(0,1)$ but g' is not bounded on $(0,1)$. This by itself does not imply that $g \notin BV[0,1]$, since being Lipschitz is only a sufficient condition. But it turns out that $g \notin BV[0,1]$. To see this, let $x_n = \sqrt{\frac{1}{(2n+1)\frac{\pi}{2}}},\ n = 0,1,2\ldots$. Then $\sum\limits_{i=1}^{n} |g(x_i) - g(x_{i-1})| \geq \sum\limits_{i=1}^{n} \frac{1}{(2i+1)\frac{\pi}{2}}$ and hence $T(g,[0,1]) = \infty$.

(f) It is known (see Royden (1988), Chapter 4) that if $f : [a,b] \to \mathbb{R}$ is nondecreasing, then f is differentiable a.e. (m) on (a,b) and $\int_{[a,b]} f' dm \leq f(b) - f(c)$, where (m) denotes the Lebesgue measure. This implies that if $f \in BV[a,b]$, then f is differentiable a.e. (m) on (a,b) and so, $\int_{[a,b]} |f'| dm \leq T(f,[a,b])$.

4.4 Absolutely continuous functions on $\mathbb{R}$

Definition 4.4.1: A function $F : \mathbb{R} \to \mathbb{R}$ is *absolutely continuous* (a.c.) if for all $\epsilon > 0$, there exists $\delta > 0$ such that if $I_j = [a_j, b_j],\ j = 1,2,\ldots,k$ ($k \in \mathbb{N}$) are disjoint and $\sum_{j=1}^{k}(b_j - a_j) < \delta$, then $\sum_{j=1}^{k} |F(b_j) - F(a_j)| < \epsilon$.

By the mean value theorem, it follows that if F is differentiable and $F'(\cdot)$ is bounded, then F is a.c. Also note that F is a.c. implies F is uniformly continuous.

Definition 4.4.2: A function $F : [a, b] \to \mathbb{R}$ is *absolutely continuous* if the function $\tilde{F}$, defined by

$$\tilde{F}(x) = \begin{cases} F(x) & \text{if} \quad a \leq x \leq b, \\ F(a) & \text{if} \quad x < a, \\ F(b) & \text{if} \quad x > b, \end{cases}$$

is absolutely continuous.

Thus $F(x) = x$ is a.c. on $\mathbb{R}$. Any polynomial is a.c. on any bounded interval but not necessarily on all of $\mathbb{R}$. For example, $F(x) = x^2$ is a.c. on any bounded interval but not a.c. on $\mathbb{R}$, since it is not uniformly continuous on $\mathbb{R}$.

The main result of this section is the following result due to H. Lebesgue, known as the *fundamental theorem of Lebesgue integral calculus.*

Theorem 4.4.1: *A function $F : [a, b] \to \mathbb{R}$ is absolutely continuous iff there is a function $f : [a, b] \to \mathbb{R}$ such that f is Lebesgue measurable and integrable w.r.t. m and such that*

$$F(x) = F(a) + \int_{[a,x]} f \, dm, \quad \textit{for all} \quad a \leq x \leq b \tag{4.1}$$

where m is the Lebesgue measure.

Proof: First consider the "if part." Suppose that (4.1) holds. Since $\int_{[a,b]} |f| dm < \infty$, for any $\epsilon > 0$, there exists a $\delta > 0$ such that (cf. Proposition 2.5.8).

$$m(A) < \delta \Rightarrow \int_A |f| dm < \epsilon. \tag{4.2}$$

Thus, if $I_j = (a_j, b_j), \subset [a, b],\ j = 1, 2, \ldots, k$ are such that $\sum_{j=1}^k (b_j - a_j) < \delta$, then

$$\sum_{j=1}^k |F(b_j) - F(a_j)| \leq \int_{\bigcup_{j=1}^k I_j} |f| dm < \epsilon,$$

since $m(\bigcup_{j=1}^k I_j) \leq \sum_{j=1}^k (b_j - a_j) < \delta$ and (4.2) holds. Thus, F is a.c.

Next consider the "only if part." It is not difficult to verify (Problem 4.18) that F a.c. implies F is of bounded variation on any finite interval $[a, b]$ and both the positive and the negative variations of F on $[a, b]$ are a.c. as well. Hence, it suffices to establish (4.1) assuming that F is a.c. and nondecreasing. Let μ_F be the Lebesgue-Stieltjes measure generated by $\tilde{F}$ as in Definition 4.4.2. It will now be shown that μ_F is absolutely continuous w.r.t. the Lebesgue measure. Fix $\epsilon > 0$. Let $\delta > 0$ be chosen so that

$$(a_j, b_j) \subset [a, b], j = 1, 2, \ldots, k, \sum_{j=1}^k (b_j - a_j) < \delta \Rightarrow \sum_{j=1}^k |F(b_j) - F(a_j)| < \epsilon.$$

Let $A \in \mathcal{M}_m$, $A \subset (a,b)$, and $m(A) = 0$. Then, there exist a countable collection of disjoint open intervals $\{I_j = (a_j, b_j) : I_j \subset [a,b]\}_{j\geq 1}$ such that

$$A \subset \bigcup_{j\geq 1} I_j \quad \text{and} \quad \sum_{j\geq 1}(b_j - a_j) < \delta.$$

Thus

$$\begin{aligned} \mu_F\Big(A \cap \bigcup_{j=1}^{k} I_j\Big) &\leq \mu_F\Big(\bigcup_{i=1}^{k} I_j\Big) \\ &\leq \sum_{j=1}^{k} \mu_F(I_j) = \sum_{j=1}^{k}(F(b_j) - F(a_j)) < \epsilon \end{aligned}$$

for all $k \in \mathbb{N}$.

Since $A \subset \bigcup_{j\geq 1} I_j$, by the m.c.f.b. property of $\mu_F(\cdot)$, $\mu_F(A) = \lim_{k\to\infty} \mu_F(A \cap \bigcup_{j=1}^{k} I_j) \leq \epsilon$. This being true for any $\epsilon > 0$, it follows that $\mu_F(A) = 0$. Since F is continuous, $\mu_F(\{a,b\}) = 0$ and hence $\mu_F\big((a,b)^c\big) = 0$. Thus, $\mu_F \ll m$, i.e., μ_F is dominated by m. Now, by the Radon-Nikodym theorem (cf. Theorem 4.1.1 (ii)), there exists a nonnegative measurable function f such that $A \in \mathcal{M}_m$ implies that $\mu_F(A) = \int_{A\cap[a,b]} f\, dm$ and, in particular, for $a \leq x \leq b$,

$$\mu_F([a,x]) = F(x) - F(a) = \int_{[a,x]} f\, dm,$$

i.e., (4.1) holds. □

The representation (4.1) of an absolutely continuous F can be strengthened as follows:

Theorem 4.4.2: *Let $F : \mathbb{R} \to \mathbb{R}$ satisfy (4.1). Then F is differentiable a.e. (m) and $F'(\cdot) = f(\cdot)$ a.e. (m).*

For a proof of this result, see Royden (1988), Chapter 4.

The relation between the notion of absolute continuity of a distribution function $F : \mathbb{R} \to \mathbb{R}$ and that of the associated Lebesgue-Stieltjes measure μ_F w.r.t. Lebesgue measure m will be discussed now.

Let $F : \mathbb{R} \to \mathbb{R}$ be a distribution function, i.e., F is nondecreasing and right continuous. Let μ_F be the associated Lebesgue-Stieltjes measure such that $\mu_F((a,b]) = F(b) - F(a)$ for all $-\infty < a < b < \infty$. Recall that F is said to be *absolutely continuous on an interval* $[a,b]$ if for each $\epsilon > 0$, there exists a $\delta > 0$ such that for any finite collection of intervals $I_j = (a_j, b_j), j = 1, 2, \ldots, n$, contained in $[a,b]$,

$$\sum_{j=1}^{n}(b_j - a_j) < \delta \quad \Rightarrow \quad \sum_{j=1}^{n}(F(b_j) - F(a_j)) < \epsilon.$$

Recall also that μ_F is *absolutely continuous w.r.t. the Lebesgue measure* m if for $A \in \mathcal{B}(\mathbb{R})$, $m(A) = 0 \Rightarrow \mu_F(A) = 0$. A natural question is that if F is absolutely continuous on every interval $[a, b] \subset \mathbb{R}$, is μ_F absolutely continuous w.r.t. m and conversely? The answer is yes.

Theorem 4.4.3: *Let $F : \mathbb{R} \to \mathbb{R}$ be a nondecreasing function and let μ_F be the associated Lebesgue-Stieltjes measure. Then F is absolutely continuous on $[a, b]$ for all $-\infty < a < b < \infty$ iff $\mu_F \ll m$ where m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.*

Proof: Suppose that $\mu_F \ll m$. Then by Theorem 4.1.1, there exists a nonnegative measurable function h such that

$$\mu_F(A) = \int_A h dm \quad \text{for all} \quad A \quad \text{in} \quad \mathcal{B}(\mathbb{R}).$$

Hence, for any $a < b$ in $\mathbb{R}$ and $a < x < b$,

$$F(x) - F(a) \equiv \mu_F((a, x]) = \int_{(a,x]} h dm.$$

This implies the absolute continuity of F on $[a, b]$ as shown in Theorem 4.4.1.

Conversely, if F is absolutely continuous on $[a, b]$ for all $-\infty < a < b < \infty$, then as shown in the proof of the "only if" part of Theorem 4.4.1, for all $-\infty < a < b < \infty$, then $\mu_F(A \cap [a, b]) = 0$ if $m(A \cap [a, b]) = 0$. Thus, if $m(A) = 0$, then for all $-\infty < a < b < \infty$, $m(A \cap [a, b]) = 0$ and hence $\mu_F(A \cap [a, b]) = 0$ and hence $\mu_F(A) = 0$, i.e., $\mu_F \ll m$. □

Recall that a measure μ on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ is a *Radon measure* if $\mu(A) < \infty$ for every bounded Borel set A. In the following, let $m(\cdot)$ denote the Lebesgue measure on $\mathbb{R}^k$.

Definition 4.4.3: A Radon measure μ on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ is *differentiable* at $x \in \mathbb{R}^k$ with *derivative* $(D\mu)(x)$ if for any $\epsilon > 0$, there is a $\delta > 0$ such that

$$\left| \frac{\mu(A)}{m(A)} - (D\mu)(x) \right| < \epsilon$$

for every open ball A such that $x \in A$ and diam. $(A) \equiv \sup\{\|x - y\| : x, y \in A\}$, the diameter of A, is less than δ.

Theorem 4.4.4: *Let μ be a Radon measure on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$. Then*

(i) μ is differentiable a.e. (m), $D\mu(\cdot)$ is Lebesgue measurable, and ≥ 0 a.e. (m) and for all bounded Borel sets $A \in \mathcal{B}(\mathbb{R}^k)$,

$$\int_A D\mu(\cdot) dm \leq \mu(A).$$

(ii) *Let* $\mu_a(A) \equiv \int_A D\mu(\cdot)dm$, $A \in \mathcal{B}(\mathbb{R}^k)$. *Let* $\mu_s(\cdot)$ *be the unique measure on* $\mathcal{B}(\mathbb{R}^k)$ *such that for all bounded Borel sets* A

$$\mu_s(A) = \mu(A) - \mu_a(A).$$

Then

$$\mu_s \perp m \quad \text{and} \quad D\mu_s(\cdot) = 0 \quad \text{a.e. } (m).$$

For a proof, see Rudin (1987).

Remark 4.4.1: By the uniqueness of the Lebesgue decomposition, it follows that a Radon measure μ on $\mathcal{B}(\mathbb{R}^k)$ is $\perp m$ iff $D\mu(\cdot) = 0$ a.e. (m) and is $\ll m$ iff $\mu(A) = \int_A D\mu(\cdot)dm$ for all $A \in \mathcal{B}(\mathbb{R}^k)$.

Let $f : \mathbb{R}^k \to \mathbb{R}_+$ be integrable w.r.t. m on bounded sets. Let $\mu(A) \equiv \int_A f dm$ for $A \in \mathcal{B}(\mathbb{R}^k)$. Then $\mu(\cdot)$ is a Radon measure and that is $\ll m$ and hence by Theorem 4.4.4

$$D\mu(x) = f(x) \quad \text{for almost all} \quad x(m).$$

That is, for almost all $x(m)$, for each $\epsilon > 0$, there is a $\delta > 0$ such that

$$\Big|\frac{1}{m(A)} \int_A f dm - f(x)\Big| < \epsilon$$

for all open balls A such that $x \in A$ and diam. $(A) < \delta$.

It turns out that a stronger result holds.

Theorem 4.4.5: *For almost all* $x(m)$, *for each* $\epsilon > 0$, *there is a* $\delta > 0$ *such that*

$$\frac{1}{m(A)} \int_A |f - f(x)| dm < \epsilon$$

for all open balls A *such that* $x \in A$ *and diam.* $(A) < \delta$ *(see Problems 4.23, 4.24).*

Theorem 4.4.6: (*Change of variables formula in* $\mathbb{R}^k$, $k > 1$). *Let* V *be an open set in* $\mathbb{R}^k$. *Let* $T \equiv (T_1, T_2, \ldots, T_k)$ *be a map from* $\mathbb{R}^k \to \mathbb{R}^k$ *such that for each* i, $T_i : \mathbb{R}^k \to \mathbb{R}$ *and* $\frac{\partial T_i(\cdot)}{\partial x_j}$ *exists on* V *for all* $1 \le i, j \le k$. *Suppose that the Jacobian* $J_T(\cdot) \equiv det\big(\big(\frac{\partial T_i(\cdot)}{\partial x_j}\big)\big)$ *is continuous and positive on* V. *Suppose further that* $T(V)$ *is a bounded open set* W *in* $\mathbb{R}^k$ *and that* T *is* $(1-1)$ *and* T^{-1} *is continuous. Then*

(i) *For all Borel set* $E \subset V$, $T(E)$ *is a Borel set* $\subset W$.

(ii) $\nu(\cdot) \equiv m(T(\cdot))$ *is a measure on* $\mathcal{B}(W)$ *and* $\nu \ll m$ *with*

$$\frac{d\nu(\cdot)}{dm} = J_T(\cdot).$$

(iii) For any $h \in L^1(W, m)$

$$\int_W h dm = \int_V h\big(T(\cdot)\big) J_T(\cdot) dm.$$

(iv) $\lambda(\cdot) \equiv mT^{-1}(\cdot)$ *is a measure on* $\mathcal{B}(W)$ *and* $\lambda \ll m$ *with*

$$\frac{d\lambda}{dm} = |J\big(T^{-1}(\cdot)\big)|^{-1}.$$

(v) For any $\mu \ll m$ *on* $\mathcal{B}(V)$ *the measure* $\psi(\cdot) \equiv \mu T^{-1}(\cdot)$ *is dominated by* m *with*

$$\frac{d\psi}{dm}(\cdot) = \Big(\frac{d\mu}{dm}\Big)\big(T^{-1}(\cdot)\big)\Big(J_T\big(T^{-1}(\cdot)\big)\Big)^{-1} \quad on \quad W.$$

For a proof see Rudin (1987), Chapter 7.

4.5 Singular distributions

4.5.1 Decomposition of a cdf

Recall that a *cumulative distribution function* (cdf) on $\mathbb{R}$ is a function $F : \mathbb{R} \to [0, 1]$ such that it is nondecreasing, right continuous, $F(-\infty) = 0$, $F(\infty) = 1$. In Section 2.2, it was shown that any cdf F on $\mathbb{R}$ can be written as

$$F = \alpha F_d + (1 - \alpha)F_c, \tag{5.1}$$

where F_d and F_c are discrete and continuous cdfs respectively. In this section, the cdf F_c will be further decomposed into a singular continuous and absolutely continuous cdfs.

Definition 4.5.1: A cdf F is *singular* if $F' \equiv 0$ almost everywhere w.r.t. the Lebesgue measure on $\mathbb{R}$.

Example 4.5.1: The cdfs of Binomial, Poisson, or any integer valued random variables are singular.

It is known (cf. Royden (1988), Chapter 5) that a monotone function $F : \mathbb{R} \to \mathbb{R}$ is differentiable almost everywhere w.r.t. the Lebesgue measure and its derivative F' satisfies

$$\int_a^b F'(x)dx \leq F(b) - F(a), \tag{5.2}$$

for any $-\infty < a < b < \infty$.

For $x \in \mathbb{R}$, let $\tilde{F}_{ac}(x) \equiv \int_{-\infty}^{x} F_c'(t)dt$ and $\tilde{F}_{sc}(x) \equiv F_c(x) - \tilde{F}_{ac}(x)$. If $\tilde{\beta} \equiv \int_{-\infty}^{\infty} F_c'(t)dt = \tilde{F}_{ac}(\infty) = 0$, then $F_c'(t) = 0$ a.e. and so F_c is singular continuous. If $\tilde{\beta} = 1$, then $F_c = \tilde{F}_{ac}$ and so, F_c is absolutely continuous. If $0 < \alpha < 1$ and $0 < \tilde{\beta} < 1$, then F can be written as

$$F = \alpha F_d + \beta F_{ac} + \gamma F_{sc}, \tag{5.3}$$

where $\beta = (1-\alpha)\tilde{\beta}$, $\gamma = (1-\alpha)(1-\tilde{\beta})$, $F_{ac} = \tilde{\beta}^{-1}\tilde{F}_{ac}$, $F_{sc} = (1-\tilde{\beta})^{-1}\tilde{F}_{sc}$, and F_d is as in (5.1). Note that F_d, F_{ac}, F_{sc} are all cdfs and α, β, γ are nonnegative numbers adding up to 1. Summarizing the above discussions, one has the following:

Proposition 4.5.1: *Given any cdf F, there exist nonnegative constants α, β, γ and cdfs F_d, F_{ac}, F_{sc} satisfying (a) $\alpha + \beta + \gamma = 1$, and (b) F_d is discrete, F_{ac} is absolutely continuous, F_{sc} is singular continuous, such that the decomposition (5.3) holds.*

It can be shown that the constants α, β, and γ are uniquely determined, and that when $0 < \alpha < 1$, the decomposition (5.1) is unique, and that when $0 < \alpha, \beta, \gamma < 1$, the decomposition (5.3) is unique. The decomposition (5.3) also has a probabilistic interpretation. Any random variable X can be realized as a randomized choice over three random variables X_d, X_{ac}, and X_{sc} having cdfs F_d, F_{ac}, and F_{sc}, respectively, and with corresponding randomization probabilities α, β, and γ. For more details see Problem 6.15 in Chapter 6.

4.5.2 Cantor ternary set

Recall the construction of the Cantor set from Section 1.3.

Let $I_0 = [0,1]$ denote the unit interval. If one deletes the open middle third of I_0, then one gets two disjoint closed intervals $I_{11} = \left[0, \frac{1}{3}\right]$ and $I_{12} = \left[\frac{2}{3}, 1\right]$. Proceeding similarly with the closed intervals I_{11} and I_{12}, one gets four disjoint intervals $I_{21} = \left[0, \frac{1}{9}\right]$, $I_{22} = \left[\frac{2}{9}, \frac{1}{3}\right]$, $I_{23} = \left[\frac{2}{3}, \frac{7}{9}\right]$, $I_{24} = \left[\frac{8}{9}, 1\right]$, and so on. Thus, at each step, deleting the open middle third of the closed intervals constructed in the previous step, one is left with 2^n disjoint closed intervals each of length 3^{-n} after n steps. Let $C_n = \bigcup_{j=1}^{2^n} I_{nj}$ and $C = \bigcap_{n=1}^{\infty} C_n$. By construction $C_{n+1} \subset C_n$ for each n and C_n's are closed sets. With $m(\cdot)$ denoting Lebesgue measure, one has $m(C_0) = 1$ and $m(C_n) = 2^n 3^{-n} = \left(\frac{2}{3}\right)^n$.

Definition 4.5.2: The set $C \equiv \bigcap_{n=1}^{\infty} C_n$ is called the *Cantor ternary* set or simply the *Cantor set*.

Since $m(C_0) = 1$, by m.c.f.a. $m(C) = \lim_{n\to\infty} m(C_n) = \lim_{n\to\infty} \left(\frac{2}{3}\right)^n = 0$. Thus, the Cantor set C has zero Lebesgue measure. Next, let $U_1 = U_{11} = \left(\frac{1}{3}, \frac{2}{3}\right)$ be the deleted interval at the first stage, $U_2 = U_{21} \cup U_{22} =$

$\left(\frac{1}{9}, \frac{2}{9}\right) \cup \left(\frac{7}{9}, \frac{8}{9}\right)$ be the union of the deleted intervals at the second stage, and similarly, $U_n = \bigcup_{j=1}^{2^{n-1}} U_{nj}$ at stage n. Thus $C^c = U = \bigcup_{n=1}^{\infty} \left(\bigcup_{j=1}^{2^{n-1}} U_{nj}\right)$ is open and $m(C^c) = 1$. Since $C \cup C^c = [0,1]$ and C^c is open, it follows that C is nonempty. In fact, C is uncountably infinite as will be shown now. To do this, one needs the concept of *p-nary expansion of numbers* in [0,1]. Fix a positive integer $p > 1$. For each x in [0,1), let $a_1(x) = \lfloor px \rfloor$ where $\lfloor t \rfloor = n$ if $n \le t < n+1$. Thus $a_1(x) \le px < a_1(x) + 1$ and $a_1(x) \in \{0, 1, \ldots, p-1\}$, i.e., $\frac{a_1(x)}{p} \le x < \frac{a_1(x)}{p} + \frac{1}{p}$. Thus, if $\frac{k}{p} \le x < \frac{k+1}{p}$ for some $k = 0, 1, 2, \ldots, p-1$, then $a_1(x) = k$. Next, let $x_1 \equiv x - \frac{a_1(x)}{p}$ and $a_2(x) = \lfloor p^2 x_1 \rfloor$. Then, $x_1 \in \left[0, \frac{1}{p}\right)$ and

$$\frac{a_2(x)}{p^2} \le x_1 = x - \frac{a_1(x)}{p} < \frac{a_2(x)}{p^2} + \frac{1}{p^2}$$

and $a_2(x) \in \{0, 1, 2, p-1\}$. Next, let $0 \le x_2 \equiv x - \frac{a_1(x)}{p} - \frac{a_2(x)}{p^2} < \frac{1}{p^2}$ and $a_3(x) = \lfloor p^3 x_2 \rfloor$ and so on. After k such iterations one gets

$$0 \le x - \sum_{i=1}^{k} \frac{a_i(x)}{p^i} < \frac{1}{p^k}$$

where $a_i(x) \in \{0, 1, 2, \ldots, p-1\}$ for all i. Since $\frac{1}{p^k} \to 0$ as $k \to \infty$, one gets the p-nary expansion of x in [0,1) as

$$x = \sum_{i=1}^{\infty} \frac{a_i(x)}{p^i}, \;\; a_i(x) \in \{0, 1, 2, \ldots, p-1\}. \tag{5.4}$$

Notice that if $x = \frac{k}{p}$ for some $k \in \{0, 1, 2, \ldots, p-1\}$, then in the above expansion $a_1(x) = k$ and $a_i(x) = 0$ for $i \ge 2$, and the expansion *terminates*. But since $\sum_{i=1}^{\infty} \frac{(p-1)}{p^i} = 1$, one may also write $x = \frac{k}{p} = \frac{k-1}{p} + \sum_{i=1}^{\infty} \frac{(p-1)}{p^i}$, this being an expansion which is *nonterminating* and *recurring*. It can be shown that for all x in [0,1) of the form $x = \frac{\ell}{p^m}$ for some positive integers ℓ and m, there are exactly two expansions such that one terminates and the other is nonterminating and recurring. For all other x in [0,1), the p-nary expansion is nonterminating and nonrecurring and is unique.

The *decimal expansion* corresponds with the case $p = 10$, the *binary expansion* with the case $p = 2$, and the *ternary expansion* with the case $p = 3$. Here the convention of choosing only a nonterminating expansion for each x in [0,1) is used. Thus, for example, for $p = 3$, $\frac{1}{3}$ will be replaced by $\sum_{i=2}^{\infty} \frac{2}{3^i}$ so that $a_1\left(\frac{1}{3}\right) = 0$, $a_i\left(\frac{1}{3}\right) = 2$ for $i \ge 2$. Similarly, for $p = 3$, $x = \frac{7}{9} = \frac{2}{3} + \frac{1}{9}$ will be replaced by $\frac{2}{3} + \frac{0}{3^2} + \sum_{i=3}^{\infty} \frac{2}{3^i}$ so that $a_1\left(\frac{7}{9}\right) = 2$, $a_2\left(\frac{7}{9}\right) = 0$, $a_i\left(\frac{7}{9}\right) = 2$ for $i \ge 3$. By taking $p = 2$, i.e., the binary expansion, it is seen that every x in [0,1) can be uniquely represented as $x = \sum_{i=1}^{\infty} \frac{\delta_i(x)}{2^i}$

where $\delta_i(x) \in \{0,1\}$ for all i. Thus, the interval [0,1) is in one-to-one correspondence with the set of all sequences of 0's and 1's.

It is not difficult to prove the following (Problem 4.31).

Theorem 4.5.2: *A number x belongs to the Cantor set C iff in (5.4), for all $i \geq 1$, $a_i(x)$ is either 0 or 2.*

Corollary 4.5.3: *The Cantor set C is in one-to-one correspondence with the set of all sequences of 0's and 1's and hence is in one-to-one correspondence with the unit interval [0,1].*

Remark 4.5.1: Thus the Cantor ternary set C is a closed subset of [0,1], its Lebesgue measure $m(C) = 0$, and its cardinality is the same as that of [0,1]. Further, it is *nowhere dense*, i.e., its complement U is dense in the sense that for every open interval $(a, b) \subset [0, 1]$, $U \cap (a, b)$ is nonempty. It is also possible to get a Cantor like set C_α with (Lebesgue) measure α, $0 < \alpha < 1$, by following the above iterative procedure of deleting at each stage intervals of length that is a fraction $\frac{(1-\alpha)}{3}$ of the full interval (Problem 4.32).

4.5.3 Cantor ternary function

The *Cantor ternary function* $F : [0, 1] \to [0, 1]$ is defined as follows: For $n \geq 1$, let $\{U_{nj} : j = 1, \ldots, 2^{n-1}\}$ denote the set of "deleted" intervals at step n in the definition of the Cantor set C. Define F on $C^c = U$ by

$$\begin{aligned} F(x) &= \frac{1}{2} \quad \text{on} \quad U_{11} = \left(\frac{1}{3}, \frac{2}{3}\right) \quad \text{and} \\ &= \frac{1}{4} \quad \text{on} \quad U_{21} = \left(\frac{1}{9}, \frac{2}{9}\right) \\ &= \frac{3}{4} \quad \text{on} \quad U_{22} = \left(\frac{7}{9}, \frac{8}{9}\right) \end{aligned}$$

and so on. It can be checked that F is uniformly continuous on U and has a continuous extension to $I_0 = [0, 1]$. The extension of the function F (also denoted by F) maps [0,1] onto [0,1] and is continuous and nondecreasing. Further, on U, it is differentiable with derivative $F' \equiv 0$. Since $m(U^c) = 0$, F is a singular cdf (cf. see Definition 4.5.1). It can be shown that if $\sum_{i=1}^{\infty} \frac{a_i(x)}{3^i}$ is the ternary expansion of $x \in (0, 1)$, then

$$F(x) = \sum_{i=1}^{N(x)-1} \frac{a_i(x)}{2^{i+1}} + \frac{1}{2^{N(x)}} \tag{5.5}$$

Where $N(x) = \inf\{i : i \geq 1, a_i(x) = 1\}$. For example, $x = \frac{1}{3} = \sum_{i=2}^{\infty} \frac{2}{3^i} \Rightarrow N(x) = \infty$, $a_1(x) = 0$, $a_i(x) = 2$ for $i \geq 2 \Rightarrow F(x) = \sum_{i=2}^{\infty} \frac{1}{2^i} = \frac{1}{2}$ while $x = \frac{4}{9} = \frac{1}{3} + \frac{0}{3^2} + \sum_{i=3}^{\infty} \frac{2}{3^i} \Rightarrow N(x) = 1 \Rightarrow F(x) = \frac{1}{2}$.

It can be shown that if $\{\delta_n\}_{n\geq 1}$ is a sequence of independent $\{0,1\}$ valued random variables with $P(\delta_1 = 0) = \frac{1}{2} = P(\delta_1 = 1)$, then the above F is the cdf of the random variable $X = \sum_{i=1}^{\infty} \frac{2\delta_i}{3^i}$ which lies in the Cantor set w.p. 1.

4.6 Problems

4.1 Show that in the proof of Theorem 4.1.1, $\mu(A_3) = 0$ where $A_3 = \{g \notin [0,1]\}$ and g satisfies (1.9).

(**Hint:** Apply (1.9) separately to $A_{31} = \{g > 1\}$ and $A_{32} = \{g < 0\}$.)

4.2 Verify that μ_{1s}, defined in (1.14) and μ_2 are singular.

(**Hint:** For each n, by Case 1, there exists a g_n in $L^2(D_n, \mathcal{F}_n, \mu^{(n)})$ where $\mu^{(n)} = \mu_1^{(n)} + \mu_2^{(n)}$ and $\mathcal{F}_n \equiv \{A \cap D_n : A \in \mathcal{F}\}$, such that $0 \leq \int_A g_n d\mu^{(n)} \leq \mu^{(n)}(A)$ for all A in $\mathcal{F}_n$. Let $A_{1n} = \{w : w \in D_n, 0 \leq g_n(w) < 1\}$, $A_{2n} = \{w : w \in D_n, g_n(w) = 1\}$, and $A_2 = \bigcup_{n\geq 1} A_{2n}$. Show that $\mu_2(A_2) = 0$ and $\mu_{1s}(A) = \sum_{n=1}^{\infty} \mu_{1n}(A \cap A_{2n})$ and hence $\mu_{1s}(A_2^c) = 0$.)

4.3 Let ν be the counting measure on $[0,1]$ and $\int_{[0,1]} h d\nu < \infty$ for some nonnegative function h. Show that $B = \{x : h(x) > 0\}$ is countable.

(**Hint:** Let $B_n = \{x : h(x) > \frac{1}{n}\}$. Show that B_n is a finite set for each $n \in \mathbb{N}$.)

4.4 Prove Proposition 4.1.2.

4.5 Find the Lebesgue decomposition of μ w.r.t. ν and the Radon-Nikodym derivative $\frac{d\mu_a}{d\nu}$ in the following cases where μ_a is the absolutely continuous component of μ w.r.t. ν.

(a) $\mu = N(0,1)$, $\nu =$ Exponential(1)

(b) $\mu =$ Exponential(1), $\nu = N(0,1)$

(c) $\mu = \mu_1 + \mu_2$, where $\mu_1 = N(0,1)$, $\mu_2 =$ Poisson(1) and $\nu =$ Cauchy(0, 1).

(d) $\mu = \mu_1 + \mu_2$, $\nu =$ Geometric(p), $0 < p < 1$, where $\mu_1 = N(0,1)$ and $\mu_2 =$ Poisson(1).

(e) $\mu = \mu_1 + \mu_2$, $\nu = \nu_1 + \nu_2$ where $\mu_1 = N(0,1)$, $\mu_2 =$ Poisson(1), $\nu_1 =$ Cauchy(0, 1) and $\nu_2 =$ Geometric(p), $0 < p < 1$.

(f) $\mu =$ Binomial (10, 1/2), $\nu =$ Poisson (1).

The measures referred to above are defined in Tables 4.6.1 and 4.6.2, given at the end of this section.

4.6 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $f \in L^1(\Omega, \mathcal{F}, \mu)$. Let

$$\nu_f(A) \equiv \int_A f d\mu \quad \text{for all} \quad A \in \mathcal{F}.$$

(a) Show that ν_f is a finite signed measure.

(b) Show that $\|\nu\| = \int_\Omega |f| d\mu$ and for $A \in \mathcal{F}$, $\nu_f^+(A) = \int_A f^+ d\mu$, $\nu_f^-(A) = \int_A f d\mu$, and $|\nu_f|(A) = \int_A |f|(d\mu)$.

4.7 (a) Let μ_1 and μ_2 be two finite measures such that both are dominated by a σ-finite measure ν. Show that the total variation measure of the signed measure $\mu \equiv \mu_1 - \mu_2$ is given by $|\mu|(A) = \int_A |h_1 - h_2| d\nu$ where for $i = 1, 2$, $h_i = \frac{d\mu_i}{d\nu}$.

(b) Conclude that if μ_1 and μ_2 are two measures on a countable set $\Omega \equiv \{\omega_i\}_{i \geq 1}$ with $\mathcal{F} \equiv \mathcal{P}(\Omega)$, then $|\mu|(A) = \sum_{i \in A} |\mu_1(\omega_i) - \mu_2(\omega_i)|$.

(c) Show that if μ_n is the Binomial (n, p_n) measure and μ is the Poisson (λ) measure, $0 < \lambda < \infty$, then as $n \to \infty$, $|\mu_n - \mu|(\cdot) \to 0$ uniformly on $\mathcal{P}(\mathbb{Z}_+)$ iff $np_n \to \lambda$.

(**Hint:** Show that for each $i \in \mathbb{Z}_+ \equiv \{0, 1, 2, \ldots\}$, $\mu_n(\{i\}) \to \mu(\{i\})$ and use Scheffe's theorem.)

4.8 Let ν be a finite signed measure on a measurable space $(\Omega, \mathcal{F})$ and let $|\nu|$ be the total variation measure corresponding to ν. Show that for any $B \in \Omega^+$, $B \subset \mathcal{F}$,

$$|\nu|(B) = \nu(B),$$

where Ω^+ is as defined in (2.13).

(**Hint:** For any set $A \subset \Omega^+$,

$$\nu(A) = \int_A h d|\nu| = \int_{A \cap \Omega^+} h d|\nu| \geq 0.)$$

4.9 Show that the Banach space $\mathbb{S}$ of finite signed measures on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$ is isomorphic to ℓ_1, the Banach space of absolutely convergent sequences $\{x_n\}_{n \geq 1}$ in $\mathbb{R}$.

4.10 Let μ_1 and μ_2 be two probability measures on $(\Omega, \mathcal{F})$.

(a) Show that

$$\|\mu_1 - \mu_2\| = 2 \sup\{|\mu_1(A) - \mu_2(A)| : A \in \mathcal{F}\}.$$

(**Hint:** For any $A \in \mathcal{F}$, $\{A, A^c\}$ is a partition of Ω and so $\|\mu_1 - \mu_2\| \geq |\mu_1(A) - \mu_2(A)| + |\mu_1(A^c) - \mu_2(A^c)| = 2|\mu_1(A) - \mu_2(A)|$, since μ_1 and μ_2 are probability measures. For the opposite inequality, use the Hahn decomposition of Ω w.r.t. $\mu_1 - \mu_2$ and the fact $\|\mu_1 - \mu_2\| = |(\mu_1 - \mu_2)(\Omega^+)| + |(\mu_1 - \mu_2)(\Omega^-)|$.)

(b) Show that $\|\mu_1 - \mu_2\|$ is also equal to

$$\sup\left\{\left|\int f d\mu_1 - \int f d\mu_2\right| : f \in B(\Omega, \mathbb{R})\right\}$$

where $B(\Omega, \mathbb{R})$ is the collection of all $\mathcal{F}$-measurable functions from Ω to $\mathbb{R}$ such that $\sup\{|f(\omega)| : \omega \in \Omega\} \leq 1$.

4.11 Let $(\Omega, \mathcal{F})$ be a measurable space.

(a) Let $\{\mu_n\}_{n\geq 1}$ be a sequence of finite measures on $(\Omega, \mathcal{F})$. Show that there exists a probability measure λ such that $\mu_n \ll \lambda$.

(**Hint:** Consider $\lambda(\cdot) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{\mu_n(\cdot)}{\mu_n(\Omega)}$.)

(b) Extend (a) to the case where $\{\mu_n\}_{n\geq 1}$ are σ-finite.

(c) Conclude that for any sequence $\{\nu_n\}_{n\geq 1}$ of finite signed measures on $(\Omega, \mathcal{F})$, there exists a probability measure λ such that $|\nu_n| \ll \lambda$ for all $n \geq 1$.

4.12 Let $\{\mu_n\}_{n\geq 1}$ be a sequence of finite measures on a measurable space $(\Omega, \mathcal{F})$. Show that there exists a finite measure μ on $(\Omega, \mathcal{F})$ such that $\|\mu_n - \mu\| \to 0$ iff there is a finite measure λ dominating μ and μ_n, $n \geq 1$ such that the Radon-Nikodym derivatives $f_n \equiv \frac{d\mu_n}{d\lambda} \to f \equiv \frac{d\mu}{d\lambda}$ in measure on $(\Omega, \mathcal{F}, \lambda)$ and $\mu_n(\Omega) \to \mu(\Omega)$.

4.13 (a) Let μ_1 and μ_2 be two finite measures on $(\Omega, \mathcal{F})$. Let $\mu_1 = \mu_{1a} + \mu_{1s}$ be the Lebesgue-Radon-Nikodym decomposition of μ_1 w.r.t. μ_2 as in Theorem 4.1.1. Show that if $\mu = \mu_1 - \mu_2$, then for all $A \in \mathcal{F}$,

$$|\mu|(A) = \int_A |h - 1| d\mu_2 + \mu_{1s}(A) \quad \text{where} \quad h = \frac{d\mu_{1a}}{d\mu_2}$$

is the Radon-Nikodym derivative of μ_{1a} w.r.t. μ_2. Conclude that if $\mu_1 \perp \mu_2$, then $|\mu|(\cdot) = \mu_1(\cdot) + \mu_2(\cdot)$ and if $\mu_1 \ll \mu_2$, then $|\mu|(A) = \int_A \left|\frac{d\mu_1}{d\mu_2} - 1\right| d\mu_2$.

(b) Compute $|\mu|(\cdot)$, $\|\mu\|$ if $\mu = \mu_1 - \mu_2$ for the following cases

(i) $\mu_1 = N(0, 1)$, $\mu_2 = N(1, 1)$

(ii) $\mu_1 =$ Cauchy (0,1), $\mu_2 = N(0, 1)$

(iii) $\mu_1 = N(0, 1)$, $\mu_2 =$ Poisson (λ).

(c) Establish Proposition 4.2.7.

4.14 Give another proof of the completeness of $(\mathbb{S}, \|\cdot\|))$ by verifying the following steps.

(a) For any sequence $\{\nu_n\}_{n\geq 1}$ in $\mathbb{S}$, there is a finite measure λ and $\{f_n\}_{n\geq 1} \subset L^1(\Omega, \mathcal{F}, \lambda)$ such that

$$\nu_n(A) = \int_A f_n d\lambda \quad \text{for all} \quad A \in \mathcal{F}, \quad \text{for all} \quad n \geq 1.$$

(b) $\{\nu_n\}_{n\geq 1}$ Cauchy in $\mathbb{S}$ is the same as $\{f_n\}_{n\geq 1}$ Cauchy in $L^1(\Omega, \mathcal{F}, \lambda)$ and hence, the completeness of $(\mathbb{S}, \|\cdot\|))$ follows from the completeness of $L^1(\Omega, \mathcal{F}, \lambda)$.

4.15 Let $f, g \in BV[a, b]$.

(a) Show that $P(f + g; [a, b]) \leq P(f; [a, b]) + P(g; [a, b])$ and that the same is true for $N(\cdot; \cdot)$ and $T(\cdot; \cdot)$.

(b) Show that for any $c \in \mathbb{R}$,

$$P(cf; [a, b]) = |c|P(f; [a, b])$$

and do the same for $N(\cdot; \cdot)$ and $T(\cdot; \cdot)$.

(c) For any $a < c < b$, $P(f; [a, b]) = P(f; [a, c]) + P(f; [c, b])$.

4.16 Let $\{f_n\}_{n\geq 1} \subset BV[a, b]$ and let $\lim_n f_n(x) = f(x)$ for all x in $[a, b]$. Show that $P(f; [a, b]) \leq \lim_{n\to\infty} P(f_n; [a, b])$ and do the same for $N(\cdot; \cdot)$ and $T(\cdot; \cdot)$.

4.17 Let $f \in BV[a, b]$. Show that f is continuous except on an at most countable set.

4.18 Let $F : [a, b] \to \mathbb{R}$ be a.c. Show that it is of bounded variation.

(**Hint:** By the definition of a.c., for $\epsilon = 1$, there is a $\delta_1 > 0$ such that $\sum_{j=1}^{k} |a_j - b_j| < \delta_1 \Rightarrow \sum_{j=1}^{k} |F(a_j) - F(b_j)| < 1$. Let M be an integer $> \frac{b-a}{\delta} + 1$. Show that $T(F, [a, b]) \leq M$.)

4.19 Let F be an absolutely continuous nondecreasing function on $\mathbb{R}$. Let μ_F be the Lebesgue-Stieltjes measure corresponding to F. Show that for any $h \in L^1(\mathbb{R}, \mathcal{M}_{\mu_F}, \mu_F)$,

$$\int_{\mathbb{R}} h d\mu_F = \int hf dm$$

where f is a nonnegative measurable function such that $F(b) - F(a) = \int_{[a,b]} f dm$ for any $a < b$.

4.20 Let $F : [a,b] \to \mathbb{R}$ be absolutely continuous with $F'(\cdot) > 0$ a.e. on $[a,b]$, where $-\infty < a < b < \infty$. Let $F(a) = c$ and $F(b) = d$. Let $m(\cdot)$ denote the Lebesgue measure on $\mathbb{R}$. Show the following:

(a) (*Change of variables formula*). For any $g : [c,d] \to \mathbb{R}$ and Lebesgue measurable and integrable w.r.t. m

$$\int_{[c,d]} g dm = \int_{[a,b]} g(F) F' dm.$$

(b) For any Borel set $E \subset [a,b]$, $F(E)$ is also a Borel set.

(c) $\nu(\cdot) \equiv m\big(F(\cdot)\big)$ is a measure on $\mathcal{B}([a,b])$ and $\nu \ll m$ with

$$\frac{d\nu}{dm}(\cdot) = F'(\cdot).$$

(d) $\lambda(\cdot) \equiv mF^{-1}(\cdot)$ is a measure on $\mathcal{B}([c,d])$ and $\lambda \ll m$ with

$$\frac{d\lambda}{dm}(\cdot) = \Big(F'\big(F^{-1}(\cdot)\big)\Big)^{-1}.$$

(e) For any measure $\mu \ll m$ on $\mathcal{B}([a,b])$ the measure $\psi(\cdot) \equiv \mu F^{-1}(\cdot)$ is dominated by m with

$$\frac{d\psi}{dm} = \frac{d\mu}{dm}\big(F^{-1}(\cdot)\big)\Big(F'\big(F^{-1}(\cdot)\big)\Big)^{-1}.$$

(f) Establish (a) assuming that g and F' are both continuous noting that both integrals reduce to Riemann integrals.

(**Hint:**

(i) Verify (a) for $g = I_{[a,b]}$, $c < \alpha < \beta < d$ and approximate by step functions.

(ii) Show that F is $(1-1)$ and $F^{-1}(\cdot)$ is continuous and hence Borel measurable.

(iii) Show that $\nu(\cdot) = \mu_F$, the Lebesgue-Stieltjes measure corresponding to F.

(iv) Use the fact that for any $c \le \alpha \le \beta \le d$,

$$\begin{aligned}
\psi([\alpha,\beta]) &= \mu([\gamma,\delta]), \quad \text{where} \quad \gamma = F^{-1}(\alpha),\ \delta = F^{-1}(\beta), \\
&= \int_{[\gamma,\delta]} g dm, \quad \text{where} \quad g = \frac{d\mu}{dm} \\
&= \int_{[\gamma,\delta]} \frac{g\Big(F^{-1}\big(F(\cdot)\big)\Big)}{F'\Big(F^{-1}\big(F(\cdot)\big)\Big)} F'(\cdot) dm \\
&= \int_{[\alpha,\beta]} \frac{g\big(F^{-1}(\cdot)\big)}{F'\big(F^{-1}(\cdot)\big)} dm \quad \text{by (a).} \quad)
\end{aligned}$$

4.21 Let $F : \mathbb{R} \to \mathbb{R}$ be absolutely continuous on every finite interval.

(a) Show that the f in (4.1) can be chosen independently of the interval $[a, b]$.

(b) Further, if f is integrable over $\mathbb{R}$, then $\lim_{x\to-\infty} F(x) \equiv F(-\infty)$ and $\lim_{x\to\infty} F(x) \equiv F(\infty)$ exist and $F(x) = F(-\infty) + \int_{(-\infty,x)} f d\mu_L$ for all x in $\mathbb{R}$.

(c) Give an example where $F : \mathbb{R} \to \mathbb{R}$ is a.c., but f is not integrable over $\mathbb{R}$.

4.22 Let $F : \mathbb{R} \to \mathbb{R}$ be absolutely continuous on bounded intervals. Let $\{I_j = 1 \le j \le k \le \infty\}$ be a collection of disjoint intervals such that $\bigcup_{j=1}^k I_j \equiv \mathbb{R}$ and on each I_j, $F'(\cdot) > 0$ a.e. or $F'(\cdot) < 0$ a.e. w.r.t. m.

(a) Show that for any $h \in L^1(\mathbb{R}, m)$,

$$\int_{\mathbb{R}} hdm = \int_{\mathbb{R}} h\big(F(\cdot)\big)|F'(\cdot)|dm.$$

(b) Show that if μ is a measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$ dominated by m then the measure $\mu F^{-1}(\cdot)$ is also dominated by m and

$$\frac{d\mu F^{-1}}{dm}(y) = \sum_{x_j \in D(y)} \frac{f(x_j)}{|F'(x_j)|}$$

where $f(\cdot) = \frac{d\mu}{dm}$ and $D(y) = \{x_j : x_j \in I_j, F(x_j) = y\}$.

(c) Let μ be the $N(0, 1)$ measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$, i.e.,

$$\frac{d\mu}{dm}(x) = \frac{1}{\sqrt{2n}}\, e^{-\frac{x^2}{2}}, \ -\infty < x < \infty.$$

Let $F(x) = x^2$. Find $\frac{d\mu F^{-1}}{dm}$.

4.23 Let $f : \mathbb{R} \to \mathbb{R}$ be integrable w.r.t. m on bounded intervals. Show that for almost all x_0 in $\mathbb{R}$ (w.r.t. m),

$$\lim_{\substack{a\uparrow x_0\\ b\downarrow x_0}} \frac{1}{(b-a)} \int_a^b |f(x) - f(x_0)|dx = 0.$$

(**Hint:** For each rational r, by Theorem 4.4.4

$$\lim_{\substack{a\uparrow x_0\\ b\downarrow x_0}} \frac{1}{(b-a)} \int_a^b |f(x) - r|dx = |f(x_0) - r|.$$

a.e. (m). Let A_r denote the set of x_0 for which this fails to hold. Let $A = \bigcup_{r\in\mathbb{Q}} A_r$. Then $m(A) = 0$. For any $x_0 \notin A$ and any $\epsilon > 0$, choose a rational r such that $|f(x_0) - r| < \epsilon$ and now show that

$$\lim_{\substack{a\uparrow x_0\\ b\downarrow x_0}} \frac{1}{(b-a)} \int_a^b |f(x) - f(x_0)|dx < \epsilon.\ )$$

4.24 Use the hint to the above problem to establish Theorem 4.4.5.

4.25 Let $(\Omega, \mathcal{B})$ be a measurable space. Let $\{\mu_n\}_{n\geq 1}$ and μ be σ-finite measures on $(\Omega, \mathcal{B})$. Let for each $n \geq 1$, $\mu_n = \mu_{na} + \mu_{ns}$ be the Lebesgue decomposition of μ_n w.r.t. μ with $\mu_{na} \ll \mu$ and $\mu_{ns} \perp \mu$. Let $\lambda = \sum_{n\geq 1} \mu_n$, $\lambda_a = \sum_{n\geq 1} \mu_{na}$, $\lambda_s = \sum_{n\geq 1} \mu_{ns}$. Show that $\lambda_a \ll \mu$ and $\lambda_s \perp \mu$ and that $\lambda = \lambda_a + \lambda_s$ is the Lebesgue decomposition of λ w.r.t. μ.

4.26 Let $\{\mu_n\}_{n\geq 1}$ be Radon measures on $\big(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k)\big)$ and m be the Lebesgue measure on $\mathbb{R}^k$. Show that if $\lambda = \sum_{n=1}^{\infty} \mu_n$ is also a Radon measure, then

$$D\lambda = \sum_{n=1}^{\infty} D\mu_n \quad \text{a.e.} \quad (m).$$

(**Hint:** Use Theorem 4.4.4 and the uniqueness of the Lebesgue decomposition.)

4.27 Let F_n, $n \geq 1$ be a sequence of nondecreasing functions from $\mathbb{R} \to \mathbb{R}$. Let $F(x) = \sum_{n\geq 1} F_n(x) < \infty$ for all $x \in \mathbb{R}$. Show that $F(\cdot)$ is nondecreasing and

$$F'(\cdot) = \sum_{n\geq 1} F_n'(\cdot) \quad \text{a.e.} \quad (m).$$

4.28 Let E be a Lebesgue measurable set in $\mathbb{R}$. The *metric density* of E at x is defined as

$$D_E(x) \equiv \lim_{\delta\downarrow 0} \frac{m\big(E \cap (x-\delta, x+\delta)\big)}{2\delta}$$

if it exists. Show that $D_E(\cdot) = I_E(\cdot)$ a.e. m.

(**Hint:** Consider the measure $\lambda_E(\cdot) \equiv m(E \cap \cdot)$ on the Lebesgue σ-algebra. Show that $\lambda_E \ll m$ and find $\lambda_E'(\cdot)$ (cf. Definition 4.4.2).)

4.29 Let F, $G : [a,b] \to \mathbb{R}$ be both absolutely continuous. Show that $H = FG$ is also absolutely continuous on $[a,b]$ and that

$$\int_{[a,b]} FdG + \int_{[a,b]} GdF = F(b)G(b) - F(a)G(a).$$

4.30 Let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space. Fix $1 \leq p < \infty$. Let $T : L^p(\mu) \to \mathbb{R}$ be a bounded linear functional as defined in (3.2.10) (cf. Section 3.2). Complete the following outline of a proof of Theorem 3.2.3 (Riesz representation theorem).

(a) Let $\nu(A) \equiv T(I_A)$, $A \in \mathcal{F}$. Verify that $\nu(\cdot)$ is a signed measure on $(\Omega, \mathcal{F})$.

(b) Verify that $|\nu| \ll \mu$.

(c) Let $g \equiv \frac{d\nu}{d\mu}$. Show that $g \in L^q(\mu)$ where $q = \frac{p}{p-1}$, $1 < p < \infty$ and $q = \infty$ if $p = 1$.

(d) Show that $T = T_g$.

4.31 Prove Theorem 4.5.2 and Corollary 4.5.3.

4.32 For $0 < \alpha < 1$, construct a Cantor like set C_α as described in Remark 4.5.1.

4.33 Show that the Cantor ternary function can be expressed as in (5.5).

4.34 Let $(\Omega, \mathcal{F}, \mu)$ be a σ-finite measure space.

(a) Let $\mathcal{G}$ be a σ-algebra $\subset \mathcal{F}$. Let $f : \Omega \to \mathbb{R}_+$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable. Show that there exists a $g : \Omega \to \mathbb{R}_+$ that is $\langle \mathcal{G}, \mathcal{B}(\mathbb{R}) \rangle$-measurable and $\nu(A) \equiv \int_A f d\mu = \int_A g d\mu$ for all A in $\mathcal{G}$.

(**Hint:** Apply Theorem 4.1.1 (b) to the measures ν and μ restricted to $\mathcal{G}$. When μ is a probability measure, g is called the conditional expectation of f given $\mathcal{G}$ (cf. Chapter 12).)

(b) Now suppose $\mathcal{G} = \sigma\langle \{A_i\}_{i \geq 1} \rangle$ where $\{A_i\}_{i \geq 1}$ is a partition of $\Omega \subset \mathcal{F}$. Determine $g(\cdot)$ explicitly on each A_i such that $0 < \mu(A_i) < \infty$.

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^{1} p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1$, $n \in \mathbb{N}$	$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^{i-1} I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b])/(b-a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx,\ a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha,\beta)} \int_A x^{\alpha-1}(1-x)^{\beta-1} I_{(0,1)}(x) m(dx)$, where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b),\ a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R},\ \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \ \frac{\sigma^2}{\sigma^2+(x-\gamma)^2}\ m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R},\ \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2/\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R},\ \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma}\ \frac{e^{-(\log x-\gamma)^2/2\sigma^2}}{x}\ I_{(0,\infty)}(x) m(dx)$